

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 19

Status: active plan after BA4 integrated closure.

Supersedes: Revision 18 only for forward execution state; R1-R18 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: CLOSED BY BA2-T4.
- BA3 provenance/derivation/identity/lifecycle/change-revalidation: CLOSED BY BA3-T4.
- BA4 projection boundary/traceability/coverage/interpretation: CLOSED BY BA4-T3.
- **BAReferent** and **BAProposition**: ACCEPTED first-class semantic identity families.
- ThreatForge remains a case study/tooling instrument, not DDTA semantic authority.

2. BA4 closure result

BA4 closes the smallest current-scope projection contract for:

- immutable revisioned projection definitions;
- explicit eligible BA scope;
- **EXHAUSTIVE_FOR_DECLARED_SCOPE** | **SELECTIVE** coverage semantics;
- qualification policy for current/stale/diagnostic/pending BA inputs;
- meaning-bearing projection-item trace to BA;
- role-bound trace where contribution roles differ;
- semantic-preserving shared rendering;
- explicitly downstream method-owned interpretation;
- projection-local interpretation-rule accountability;
- BA trace plus BA3 continuity as the cross-projection common anchor; and
- baseline-scoped projection rebuild without a second project lifecycle.

BA4-T3 removes independent **omissionSemantics** as redundant because omission meaning is fixed by the coverage mode.

No universal projection ontology, universal projection DSL, new BAE family, new BA2 operator or BA1-BA3 reopen is forced.

The active closure artifact is:

- `methodology/BA4_PROJECTION_BOUNDARY_TRACEABILITY_INTERPRETATION_COVERAGE_CONTRACT_R1.md`

BA4-T1/T2 artifacts remain immutable research derivation history.

3. Base Analysis status after BA4

BA0 CLOSED
 BA1 CLOSED
 BA2 CLOSED
 BA3 CLOSED
 BA4 CLOSED

BA5 NOT STARTED / NEXT

BA6 NOT STARTED

The Base Analysis milestone as a whole is not yet closed. BA5 must establish lexical/display/assistance boundaries and BA6 must perform the integrated corpus plus holdout closure regression.

4. BA5 objective - lexical vocabulary and optional assistance

BA5 defines the smallest boundary among source wording, stable BA semantic keys, projection-local display labels, authoring synonyms and optional assistance.

The phase must preserve:

```
source lexical wording
    != stable BA semantic key
    != projection-local display label
    != authoring synonym
```

The purpose is not to build a universal natural-language lexicon. It is to prevent convenient labels, synonyms or model-generated suggestions from silently becoming semantic authority.

NLP/LLM support, if later admitted, remains candidate-producing assistance subject to provenance/review and cannot create project or BA truth by itself.

5. BA5-T1 - stable semantic key, display-label and authoring-synonym boundary trial

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA5-T1 must perform only the first lexical-boundary trial.

It must pressure-test at least:

1. one BA semantic key rendered with several human display labels without changing BA identity/meaning;
2. one source phrase that should not become the stable BA key merely because it is common in the documentation;
3. projection-local method labels that must remain distinct from BA semantic keys;
4. authoring synonyms that can help humans express existing semantics without creating new semantics;
5. ambiguity where one lexical phrase can map to more than one existing BA meaning and therefore cannot be silently normalized;
6. whether lexical metadata requires any new first-class BAE family or BA2 operator; and
7. whether a material counterexample forces the smallest BA1-BA4 reopen.

BA5-T1 guardrails

Do not:

- introduce automatic arbitrary-narrative migration;
- authorize NLP/LLM output as semantic truth;
- define a universal domain lexicon;
- convert projection-local method taxonomy into BA vocabulary;
- define AnalysisRecord/Common Finding;
- design ThreatForge classes/tables/UI;
- reopen BA0-BA4 without a material shared-semantic counterexample;
- start BA6.

BA5-T1 exit condition

Produce a falsifiable lexical responsibility lower-bound candidate and the next smallest BA5 pressure target. Do not close BA5 from one lexical trial.

6. BA6 - complete Base Analysis regression and closure

Only after BA5 closes, regress the complete BA0-BA5 design against the closed corpora and at least one structurally different holdout.

BA6 may reopen only the smallest earlier responsibility on a material counterexample.

Only BA6 may close the complete Base Analysis milestone for current thesis scope.

7. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

8. Next authorized microstep

Only after the BA4-T3 closure package is reviewed, committed, pushed and remotely verified, execute:

BA5-T1 - stable semantic key, display-label and authoring-synonym boundary trial.

Do not start optional assistance, BA6 or downstream analysis schemas before BA5-T1 is completed and reviewed.